

A Counterexample to Detlefs' Conjecture on OEIS A008365, with the Corrected Characterisation

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Abstract

OEIS A008365 lists the 13-rough numbers, that is the positive integers coprime to $2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$. In December 2011, G. Detlefs conjectured that A008365 is exactly the set of n with $n^{24} \equiv 1, 421, 631$ or $841 \pmod{2310}$. The conjecture is false. The smallest counterexample is $n = 17$, the third listed term of the sequence itself: $17^{24} \equiv 1681 \pmod{2310}$. The image of the map $n \mapsto n^{24}$ on $(\mathbb{Z}/2310\mathbb{Z})^\times$ has five elements, not four, and adjoining the missing residue 1681 makes the characterisation correct and an equivalence. We record the corrected statement, identify the counterexamples exactly as the 13-rough $n \equiv 5, 6 \pmod{11}$ (one fifth of all 13-rough numbers), and give the general statement for an arbitrary primorial and exponent, which explains both why the companion conjecture on A008364 is correct and why this one is not.

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1 The conjecture

Sequence A008365 of the On-Line Encyclopedia of Integer Sequences [1] is

$$A008365 = \{n \geq 1 : \gcd(n, 2310) = 1\} = 1, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, \dots$$

the 13-rough numbers, $2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$ being the fifth primorial. On 30 December 2011, Gary Detlefs contributed the comment

Conjecture: Numbers n such that n^{24} is congruent to $\{1, 421, 631, 841\} \pmod{2310}$.

As of 22 August 2026 the comment stands on the entry with no rebuttal. Ten days earlier, on 20 December 2011, the same contributor had proposed the companion statement for A008364, the 11-rough numbers, that they are the n with $n^6 \equiv 1$ or $169 \pmod{210}$; that one is correct and was verified on the entry by A. del Arte in January 2014 [1].

Theorem 1. *Detlefs' conjecture on A008365 is false. The integer $n = 17$ is 13-rough — it is the third term of the sequence — and*

$$17^{24} \equiv 1681 \pmod{2310}, \quad 1681 \notin \{1, 421, 631, 841\}.$$

Proof. $\gcd(17, 2310) = 1$, so 17 lies in A008365. For the residue, work prime by prime. Modulo 2, 3, 5 and 7 the exponent 24 is a multiple of the group order 1, 2, 4 and 6 respectively, so $17^{24} \equiv 1$ in each case, and therefore $17^{24} \equiv 1 \pmod{210}$. Modulo 11 we have $17 \equiv 6$ and $6^{10} \equiv 1$, so $17^{24} \equiv 6^4 = 1296 \equiv 9 \pmod{11}$. Now $1681 = 41^2$ satisfies $1681 = 8 \cdot 210 + 1 \equiv 1 \pmod{210}$ and $1681 = 152 \cdot 11 + 9 \equiv 9 \pmod{11}$, so by the Chinese remainder theorem $17^{24} \equiv 1681 \pmod{2310}$. Finally 1681 is none of 1, 421, 631, 841. $\square$

2 The corrected characterisation

Theorem 2. *For every integer n ,*

$$\gcd(n, 2310) = 1 \iff n^{24} \bmod 2310 \in \{1, 421, 631, 841, 1681\},$$

and every one of the five residues is attained, the least witnesses being $n = 1, 29, 19, 13, 17$ respectively. Detlefs' set is exactly this set with 1681 deleted.

Proof. ($\Leftarrow$) If n^{24} is congruent modulo 2310 to any of the five listed residues, then n^{24} is coprime to 2310, since each of the five is; hence so is n .

($\Rightarrow$) Let $\gcd(n, 2310) = 1$. As in Theorem 1, $n^{24} \equiv 1 \pmod{p}$ for $p = 2, 3, 5, 7$, because 24 is divisible by $p - 1$ in each case. Modulo 11 we have $n^{10} \equiv 1$, so $n^{24} \equiv n^4$, and $n^4 = (n^2)^2$ is a quadratic residue; since $\gcd(4, 10) = 2$, the image of the fourth-power map on the cyclic group $(\mathbb{Z}/11\mathbb{Z})^\times$ is precisely its subgroup of index 2, namely the quadratic residues $\{1, 3, 4, 5, 9\}$, and each value is attained. Thus $n^{24} \equiv 1 \pmod{210}$ and $n^{24} \bmod 11 \in \{1, 3, 4, 5, 9\}$. Writing the residue as $x = 1 + 210t$ with $0 \leq t \leq 10$ and using $210 \equiv 1 \pmod{11}$, we get $x \equiv 1 + t \pmod{11}$, so $t \in \{0, 2, 3, 4, 8\}$ and

$$x \in \{1, 421, 631, 841, 1681\}. \quad \square$$

Corollary 3. *A 13-rough number n is a counterexample to Detlefs' conjecture if and only if $n \equiv 5$ or $6 \pmod{11}$. These are exactly the 13-rough n with $n^4 \equiv 9 \pmod{11}$, and they form one fifth of the 13-rough numbers. The first ten are*

$$17, 61, 71, 83, 127, 137, 149, 181, 193, 247.$$

Proof. By the proof of Theorem 2, the residue 1681 occurs exactly when $n^4 \equiv 9 \pmod{11}$, and the solutions of $x^4 \equiv 9$ in $(\mathbb{Z}/11\mathbb{Z})^\times$ are $x = 5, 6$. These are 2 of the 10 classes coprime to 11, hence a proportion $1/5$ of the 13-rough numbers. The listed values are outputs of the verification script of §4. $\square$

3 Why the companion conjecture is right and this one is wrong

The two statements are instances of one elementary fact, which also shows that a characterisation of this shape exists for every primorial and every exponent — provided one takes the whole image.

Theorem 4. *Let $P = p_1 p_2 \cdots p_k$ be a product of distinct primes and let $e \geq 1$. Put*

$$H_e = \{ m^e \bmod P : \gcd(m, P) = 1 \}.$$

Then H_e is a subgroup of $(\mathbb{Z}/P\mathbb{Z})^\times$ of order

$$|H_e| = \prod_{i=1}^k \frac{p_i - 1}{\gcd(e, p_i - 1)},$$

and for every integer n one has $\gcd(n, P) = 1$ if and only if $n^e \bmod P \in H_e$.

Proof. By the Chinese remainder theorem $(\mathbb{Z}/P\mathbb{Z})^\times \cong \prod_i (\mathbb{Z}/p_i\mathbb{Z})^\times$, a product of cyclic groups of orders $p_i - 1$. In a cyclic group of order d the e -th power map is an endomorphism with kernel of order $\gcd(e, d)$, so its image is the unique subgroup of order $d / \gcd(e, d)$. Taking the product gives the stated order, and the image of a product map is the product of the images. If $\gcd(n, P) = 1$ then $n^e \bmod P$ lies in the image by definition; conversely every element of H_e is a unit, so n^e , and hence n , is coprime to P . $\square$

Corollary 5. *For $P = 210$ and $e = 6$ one gets $|H_6| = 1 \cdot 1 \cdot 2 \cdot 1 = 2$, and indeed $H_6 = \{1, 169\}$: Detlefs' A008364 conjecture is correct. For $P = 2310$ and $e = 24$ one gets $|H_{24}| = 1 \cdot 1 \cdot 1 \cdot 1 \cdot 5 = 5$, so any four-element set is too small: his A008365 conjecture cannot be correct as stated.*

Remark 6. The discrepancy is localised at the single prime 11. For $p \in \{2, 3, 5, 7\}$ one has $(p - 1) \mid 24$, so those factors contribute nothing; but $\gcd(24, 10) = 2 < 10$, so the eleventh factor contributes a subgroup of order 5. In the 210 case the only nontrivial factor is $\gcd(6, 4) = 2$ at the prime 5, giving two residues — few enough to be found by hand, which is presumably why that conjecture came out right and this one did not.

4 Computational verification

A single script (Python 3, standard library plus **sympy** for prime generation) performs the following; all checks pass.

1. $\gcd(17, 2310) = 1$ and $17^{24} \bmod 2310 = 1681 \notin \{1, 421, 631, 841\}$.
2. The full image of $n \mapsto n^{24}$ on $(\mathbb{Z}/2310\mathbb{Z})^\times$, computed by exhaustion over all 480 units, is $\{1, 421, 631, 841, 1681\}$, with least witnesses 1, 29, 19, 13, 17.

3. For all $n < 20000$, $n^{24} \bmod 2310 \in \{1, 421, 631, 841, 1681\}$ if and only if $\gcd(n, 2310) = 1$ — no exceptions.
4. Below 20000 there are 832 counterexamples to Detlefs' set; every one is congruent to 5 or 6 modulo 11, and the first ten are those listed in Corollary 3.
5. The order formula of Theorem 4 agrees with brute-force enumeration for the primorials 2, 6, 30, 210, 2310 and the exponents $e \in \{2, 4, 6, 8, 12, 24, 30, 60\}$.
6. Control: the image of $n \mapsto n^6$ on $(\mathbb{Z}/210\mathbb{Z})^\times$ is exactly $\{1, 169\}$.

Every numeral in this note is an output of that script.

5 Concluding remarks

The result is elementary and the disproof is a single modular exponentiation; we claim no difficulty for it. It is recorded because the conjecture has stood on the OEIS entry since December 2011 and because its failure is not marginal: it fails at the third term of the sequence it purports to describe, and one 13-rough number in five is a counterexample. The appropriate destination for the correction is a comment on the entry itself rather than a journal. We have found no earlier notice of the error in the literature or on the entry, but so short an observation could well have been made before.

References

- [1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequences A008364 (11-rough numbers) and A008365 (13-rough numbers); comments of G. Detlefs, 20 and 30 December 2011, and of A. del Arte, 12 January 2014.